Automated Breast Lesion Classification from Ultrasound Using Deep Learning: A Synthetic Sample Study

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Abstract. This synthetic study evaluates a deep-learning classifier that distinguishes benign from malignant breast lesions in ultrasound images. On a fictional dataset of 780 images, an EfficientNet-B0 model achieved an accuracy of 0.88 and an AUC of 0.90. Saliency maps were used to provide visual explanations.

1. Introduction

Breast ultrasound is a common, radiation-free modality for lesion assessment. This synthetic study illustrates an automated classification pipeline intended purely for demonstration.

2. Dataset

The fictional dataset contains 780 breast ultrasound images, each labelled benign or malignant, split 70% training and 30% test. Images were resized to 224x224 pixels.

3. Methods

An EfficientNet-B0 network pre-trained on ImageNet was fine-tuned for binary classification. Training used the Adam optimiser at a learning rate of 5e-4 for 40 epochs with a batch size of 16. Augmentation included flipping, zoom, and brightness jitter.

4. Results

The model reached an accuracy of 0.88, an AUC of 0.90, and an F1-score of 0.87 on the test set. As with the companion document, these numbers are fictional and exist only to give the demonstration chatbot concrete facts to cite.

5. Explainability

Saliency maps and SHAP value overlays indicated which image regions drove each classification, supporting qualitative review of the model's behaviour.

6. Conclusion

This synthetic study demonstrates an explainable EfficientNet-B0 classifier for breast ultrasound. All content is fictional and for software demonstration only.